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BANGLADESH

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STAFF REPORT FOR THE 2023 ARTICLE IV CONSULTATION, FIRST REVIEWS UNDER THE EXTENDED CREDIT FACILITY ARRANGEMENT, ARRANGEMENT UNDER THE EXTENDED FUND FACILITY, AND THE RESILIENCE AND SUSTAINABILITY FACILITY ARRANGEMENT, REQUESTS FOR A WAIVER OF NONOBSERVANCE OF A PERFORMANCE CRITERION, AND MODIFICATIONS OF PERFORMANCE CRITERIA—DEBT SUSTAINABILITY ANALYSIS

Approved By
Anne-Marie Gulde-Wolf and Boileau Yeyinou Loko (IMF) and Manuela Francisco and Mathew Verghis (IDA)

Prepared by the staffs of International Monetary Fund and the International Development Association

Bangladesh: Joint Bank Fund Debt Sustainability Analysis	
Risk of external debt distress	Low
Overall risk of debt distress	Low
Granularity in the risk rating	Tool not applicable
Application of judgement	Yes

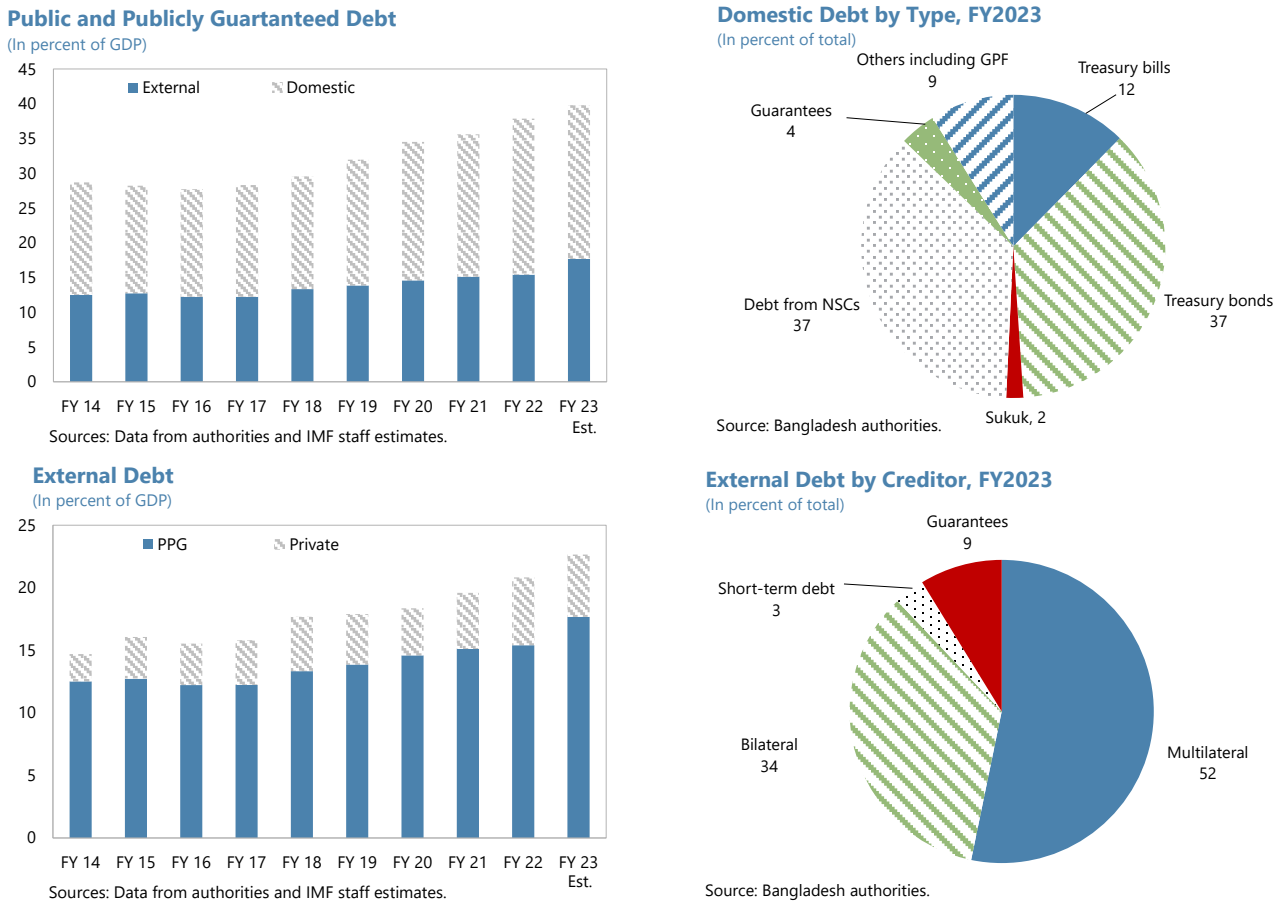
Bangladesh remains at a low risk of external and overall debt distress. Bangladesh's debt-carrying capacity is unchanged from the previous debt sustainability analysis (DSA).[1] External and overall debt indicators are below their respective thresholds under the baseline. Under a standard stress test, the present value of debt-to-export ratio temporarily breaches the threshold of external debt sustainability indicator. This short duration breach of low magnitude is discounted via judgement. Despite increased external borrowing projected in the near-term, favorable debt dynamics in the medium term keep the public and publicly guaranteed (PPG) external debt-to-GDP ratio on a declining path. Risks are tilted to the downside and include persistent inflation, increasing interest burden, revenue mobilization constraints, slowdown in major trading partners, slow implementation of macro-critical structural reforms, amplified foreign exchange (FX) pressures, elevated non-performing loans (NPLs), and climate related events.

[1] Bangladesh's Composite Index is estimated at 2.86 and is based on IMF's October 2023 WEO and WB's 2022 CPIA. The debt carrying capacity remains medium.

A. Background and Developments on Debt

1. Total public and publicly guaranteed debt in Bangladesh stood at US$166.7 billion in FY23, around 39.8 percent of GDP (Text Table 1 and Text Figure 1).[2] The majority of public debt accumulated over the last decade is domestic and denominated in local currency. In FY23, domestic debt was at 55.6 percent of PPG debt stock. Domestic debt service payments amounted to US$7.8 billion in FY23 (1.9 percent of GDP). External debt was at 44.4 percent of PPG debt stock. External debt service payments in FY23 comprised of the payments to multilateral creditor of US$2.1 billion (0.5 percent of GDP) and bilateral creditor of US$0.9 billion (around 0.2 percent of GDP). Of which, debt service payments to Paris Club and non-Paris Club creditors amounted to US$0.5 billion and US$0.4 billion, respectively.

Text Figure 1. Bangladesh: External and Domestic Debt By Type



[2] Fiscal year in Bangladesh starts in July and ends in June.

Text Table 1. Bangladesh: Decomposition of Public Debt, FY23

	Debt Stock FY2023			Debt Service FY2023		
	(In million of US$)	(Percent total debt)	(Percent GDP)	(In million of US$)	(Percent total debt)	(Percent GDP)
Total	166,655	100.0	39.8	23,325	14.0	5.6
External PPG	74,007	44.4	17.7	3,886	2.3	0.9
Medium-to-long term	71,632	43.0	17.1	3,676	2.2	0.9
Multilateral creditors[2]	38,812	23.3	9.3	2,058	1.2	0.5
IMF	1,849	1.1	0.4	270	0.2	0.1
World Bank	19,530	11.7	4.7	687	0.4	0.2
ADB	14,115	8.5	3.4	938	0.6	0.2
Other Multilaterals	3,318	2.0	0.8	162	0.1	0.0
o/w: AIIB	1,504	0.9	0.4	49	0.0	0.0
o/w: IDB	696	0.4	0.2	66	0.0	0.0
Bilateral creditors	25,350	15.2	6.1	882	0.5	0.2
Paris Club	18,327	11.0	4.4	518	0.3	0.1
o/w: Japan	11,000	6.6	2.6	145	0.1	0.0
o/w: Russia	5,900	3.5	1.4	330	0.2	0.1
Non-Paris Club	7,023	4.2	1.7	364	0.2	0.1
o/w: China	5,374	3.2	1.3	251	0.2	0.1
o/w: India	1,299	0.8	0.3	70	0.0	0.0
SOE borrowing	6,391	3.8	1.5	736	0.4	0.2
Others	1,079	0.6	0.3	NA	NA	NA
Short-term	2,375	1.4	0.6	210	0.1	0.1
Domestic	92,648	55.6	22.1	19,439	11.7	4.6
Treasury bills	11,452	6.9	2.7	413	0.2	0.1
Treasury bonds	33,897	20.3	8.1	5,957	3.6	1.4
Sukuk	1,667	1.0	0.4	85	0.1	0.0
Debt from NSCs	33,818	20.3	8.1	12,984	7.8	3.1
Guarantees	3,560	2.1	0.8	-	0.0	0.0
Others including GPF	8,254	5.0	2.0	NA	NA	NA
Memo items:						
Collateralized debt[3]	-	0.0	0.0			
Contingent liabilities	9,553	5.7	2.3			
o/w: Public guarantees	9,553	5.7	2.3			
o/w: Other explicit contingent liabilities[4]	-	0.0	0.0			
Nominal GDP[5]	418,799					

1/ As reported by Bangladesh authorities according to their classification of creditors.
2/ Multilateral creditors are simply institutions with more than one official shareholder and may not necessarily align with creditor classification under other IMF policies (e.g. Lending Into
3/ Debt is collateralized when the creditor has rights over an asset or revenue stream that would allow it, if the borrower defaults on its payment obligations, to rely on the asset or revenue
4/ Includes other-one off guarantees not included in publicly guaranteed debt (e.g. credit lines) and other explicit contingent liabilities not elsewhere classified (e.g. potential legal claims,
5/ Nominal GDP reported in this table is based on end of the period exchange rate, 106 Taka per dollar.

2. National Saving Certificates (NSCs) make up a large but declining share of domestic debt. [3] Gross NSC issuance declined 25 percent year-on-year and net issuance turned negative in FY23, as a combination of reforms initiated since 2019[4] and increases in cost-of-living reduced demand. Nonetheless, NSCs continue to make up just under 40 percent of domestic debt. Rising domestic bond yields have led to a narrowing of the gap with interest rates offered on NSCs (Text Table 2). Nonetheless, continued reforms to fully align NSC interest rates with market-determined rates, together with further liberalization of a reference lending rate (the six-month moving average

Text Table 2. Bangladesh: Selected NSCs Interest Rates

(in percent)	Three-monthly profit basis	Bangladesh bond	Family	Pensioner
Maturity	3-yr	5-yr	5-yr	5-yr
Up to Tk 1.5 million	11.0	11.3	11.5	11.8
Between Tk1.5 and Tk3 million	10.0	10.3	10.5	10.8
More than Tk 3 million	9.0	9.3	9.5	9.8
Comparable bond market yield	9.0	10.0	10.0	10.0

Source: Ministry of Finance

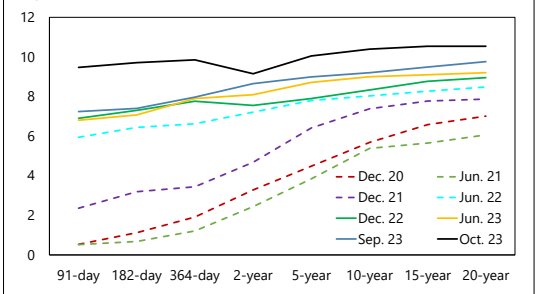
[3] NSCs were introduced in the subcontinent in 1944 by the National Savings Institute of the Ministry of Finance of India. The intent was to promote savings among the population and finance the government's budget deficit. Currently, the Department of National Savings under the Bangladesh Ministry of Finance issues NSCs.

[4] In FY2020, the government digitized the issuance of NSCs and linked it to the purchasers' tax identification number, facilitating better enforcement of existing caps, as well as increased the tax on interest income from 5 to 10 percent. This was followed by the introduction of a tiered interest rate system in September 2021 under which interest rates decline with the size of NSC holdings.

rate of T-bills, SMART), would help to improve monetary policy transmission and deepen domestic debt markets, while ultimately lowering the cost of domestic borrowing. The last Sukuk bond was issued in April 2022. Bangladesh began issuing shariah-compliant domestic investment instruments, or Sukuk bonds, in December 2020. However, amid a liquidity squeeze among Islamic banks, no new Sukuk bonds were issued in FY23 and total Sukuk bond issuance remained unchanged at 180 billion Taka. The authorities are planning to issue more Sukuk bonds in the future.

3. Yields on government securities continued to climb in FY23 and into FY24. Monetary policy tightening, exchange rate depreciation, and unsterilized FX sales contributed to tight financial conditions. Reflecting implicit interest rate caps, upheld by substantial devolvement of government securities to Bangladesh Bank (BB) below market-clearing interest rates until August 2023, yields at the short end rose faster than at the long end (Text Figure 2). However, following a recent 75 basis point hike in the policy rate and an increase in the margin on SMART of 50 basis points, long-term yields also rose. Net domestic borrowing of the central government increased by 39.1 percent in FY23 compared to FY22, primarily filled by higher net borrowing from banks and non-bank institutions as external financing and net NSC issuance declined.

Text Figure 2. Bangladesh: Government T-Bill and T-Bond Rates
(In percet)



Source: Bangladesh Bank.

4. External PPG debt stood at US$74 billion, accounting for about 17.7 percent of GDP, in FY23. External PPG debt is predominantly owed by the central government to multilateral and bilateral creditors, at about 52 percent and 34 percent of outstanding external PPG debt respectively, with the rest being short-term, others (sovereign bonds held by non-resident Bangladeshis), and guaranteed SOE debt. External project financing disbursements in FY23 were lower than projected previously. Global inflationary environment and supply chain disruptions, and inability of banks to open letters of credits (LCs) due to FX shortages, contributed to increased import costs and delayed project execution, reducing disbursements from donor partners. Private sector debt was 5 percent of GDP in FY23, declining marginally from FY23.

B. Debt Coverage

5. Debt coverage used for the analysis appropriately captures Bangladesh's debt vulnerabilities.[5] The DSA covers the full stock of public debt issued by the central government, as well as debt issued by SOEs and guaranteed by the central government. IMF financial support is included in the public debt, which is provided to the Bangladesh Bank (BB) and then on lent to the central government. Local governments are not allowed to borrow and are excluded from the analysis. The authorities are working to standardize the reporting of SOE debt to cover non-

[5] Based on the authorities' data, non-guaranteed SOE debt (outside debt coverage) is assessed to be lower than 2 percent of GDP (informing the calibrations of the contingent liability shock) and does not represent a fiscal risk based on the information available.

guaranteed debt. This DSA is prepared on a currency basis as data are not available for the residency basis. The difference between the two definitions should not materially affect the assessment.

Text Table 3. Bangladesh: Debt Coverage

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

1	The country's coverage of public debt	The central government, central bank, government-guaranteed debt		
		Default		Used for the analysis
2	Other elements of the general government not captured in 1.	0	percent of GDP	0.0
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2	percent of GDP	2.0
4	PPP	35	percent of PPP stock	0.0
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5	percent of GDP	5.0
	Total (2+3+4+5) (in percent of GDP)			7.0

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

C. Macroeconomic and Financing Assumptions

6. The macroeconomic assumptions underlying this DSA are as follows (Text Table 3):

- **Growth and inflation.** Bangladesh continues to deal with multi-faceted macroeconomic challenges. Amid the impacts of Russian invasion of Ukraine and global monetary tightening, inflation remained elevated and FX pressures continued, both amplified by expectations of further depreciation. Policies to compress domestic demand and monetary tightening suggest a subdued near-term growth outlook, compared to the previous DSA conducted in January 2023. Real GDP growth is projected to stabilize at 6 percent in FY24. Headline inflation averaged at 9 percent in FY23, driven by domestic food and energy price increase, pass-through from large Taka depreciations and elevated global commodity prices. Headline inflation is expected to decline to 7.9 percent in FY24 and further moderate over the medium and long term, staying anchored around 5.5 percent. Driven also by import compression, short-run projections assume muted domestic demand. Muted domestic demand, together with a tighter monetary policy stance, is expected to reduce inflationary pressures. Long-run projections, however, assume imports to rebound, and consequently, not to affect the long-term potential in the baseline. However, prolonged import compression could have negative impact on the potential. External debt dynamics under the baseline are favorable under a projected medium to long-term growth rate of around 6.5–7 percent. In the long run, growth will be mainly driven by investment and productivity improvement, both of which will benefit from holistic reforms supported by the IMF-supported program and the World Bank's Development Policy Credit. Increased female

participation in the workforce will provide an upside to growth as population growth moderates. Downside risks to growth could arise from persistent inflation, slowdown in major trading partners, amplified FX pressures, elevated non-performing loans (NPLs), and slow implementation of macro critical structural reforms. Climate related events continue to be a downside risk to growth.

Text Table 4. Bangladesh: Macroeconomic Assumptions [1/]

	FY22	FY23	FY24	FY25	FY26	FY27	FY28	Long Term Avg. 2/
		Est.				Proj.		
Real GDP growth								
Current	7.1	6.0	6.0	6.6	7.1	7.2	7.0	7.0
Previous	7.2	5.5	6.5	7.1	7.3	7.4	7.0	7.0
Inflation								
Current	6.1	9.0	7.9	6.8	5.5	5.5	5.5	5.5
Previous	6.1	8.9	6.5	5.6	5.5	5.5	5.7	5.7
Primary fiscal balance (in percent of GDP) 3/								
Current	-2.1	-2.6	-2.8	-2.3	-2.5	-2.5	-2.5	-2.5
Previous	-2.1	-3.8	-3.3	-3.3	-3.3	-3.3	-3.1	-2.9
Current account deficit (in percent of GDP)								
Current	-4.1	-0.7	-0.8	-2.7	-3.0	-3.0	-3.0	-2.5
Previous	-4.1	-3.2	-4.2	-3.5	-3.0	-3.0	-2.8	-2.7

1/ The previous DSA refers to the 2023 Joint DSA published in February 2023.
2/ Long tern average figures cover FY29-FY43 for the current DSA and FY28-FY42 for the previous DSA.
3/ Primary fiscal balance includes the grants. The difference in historical data are caused by GDP rebasing.

- **Fiscal policy.** The primary fiscal balance is projected to improve in FY24 and FY25 than in the previous DSA as the authorities maintain a tighter fiscal stance to help bring down inflation and ease the Balance of Payments (BoP) pressures. The primary deficit is projected to average 3.1 percent of GDP in the three subsequent fiscal years as inflation and BoP pressures subside and public investment picks up. Over the long term, the overall fiscal deficit is expected to be contained to 5 percent of GDP in line with the authorities' target. Under the IMF-supported program, stepped up efforts to mobilize fiscal revenues are expected to raise the revenue-to-GDP ratio by 1.7 percentage points during FY24 and FY26, with additional gradual increases totaling about 0.5 percent of GDP in revenues projected over the long term.

- **Climate policy.** Since the RSF arrangement, additional climate financing worth US$900 million from the World Bank (US$500 million) and ADB (US$400 million) has been approved in FY23. Additional financing is expected, including new budget support from Agence Française de Développement (AFD) amounting to US$580 million during FY24-FY26. The FY24 budget includes 0.7 percent of GDP in climate-related spending, similar to the FY23 allocation and historical spending.[6] Starting in FY25, the baseline scenario includes a gradual scaling up of

[6] Actual climate-related spending for FY23 is not yet available. Between FY18-22, climate spending averaged 0.6-0.7 percent of GDP. Data on climate-related spending covers twenty-five Ministries/Divisions, responsible for implementing major climate projects. Other Ministries/Divisions may have climate relevant expenditures not captured here.

climate investment of 0.3-0.8 percent of GDP annually supported by RSF and other financing catalyzed through development partners. RSF disbursements would help reduce the present value (PV) of debt and debt servicing burdens by substituting for more expensive domestic debt. The authorities do not intend to undertake additional climate-related spending, beyond what is already included in the baseline DSA, that would jeopardize risks to public and publicly guaranteed debt sustainability.

- **Current account (CA) dynamics.** After widening substantially in FY22, the CA deficit narrowed in FY23 due to FX shortages and LC margin requirements on the payments on imports, while exports remained resilient despite slowing external demand. Remittance inflows grew moderately by around 3 percent (y/y) in FY23, stabilizing below its historical average growth despite significant increase in migrant outflows in FY23. A reversal in the financial account—reflecting delays in repatriation and repayment of export proceeds, faster than anticipated global monetary tightening, uncertainty over FX setting, and expectation of future depreciation, continued to exert significant pressures on the Taka and FX reserves. Going forward, import compression is projected to continue into FY24. As a result, the CA deficit is expected to remain muted at 0.8 percent of GDP in FY24. From FY25 onward, the CA deficit is expected to increase, as FX pressures ease, thus helping normalize imports. In the medium to long term, the CA deficit is expected to gradually stabilize to around 3 percent of GDP, as non-ready-made garment (RMG) exports gain market share, due to structural reforms, trade facilitation, infrastructure development, and skill enhancement, as well as steady contributions from remittances. Reserves will be boosted by additional disbursements under an IMF-supported program, as well as from the World Bank and other development partners. Reserve coverage is expected to stabilize around 4 months of prospective import coverage in the medium term.

- **Financing assumptions.** Gross public financing need is projected to stabilize around 12 percent of GDP by FY34.[7] The concessionality of debt increases in the short run reflecting additional disbursements. However, they are projected to decline over the medium term, as Bangladesh graduates from the LDC status and income levels rise. Multilateral donor financing is projected to contribute to around 50 percent of total external financing in the near term, and gradually decline to 30 percent by FY44. The remainder of the external financing is assumed to come from bilateral donors. Average nominal interest rate on external debt is 2.7 percent in FY34. [8] Reflecting higher global interest rates, external interest payment projections have been revised up relative to the previous DSA. In line with expected monetary tightening to bring down inflation in the short term and further liberalization of market lending rates over the medium term, the interest rates on domestic debt have been revised up, averaging 2.2 percent in real terms over the projection period, compared to 0 percent in the previous DSA. In nominal terms, the interest rates on domestic debt are assumed to vary from 8.5 percent for T-bills, 10 percent

[7] Compared to the previous DSA, higher old domestic debt's interest payments and amortization increased GFNs by 2-3 percentage points of GDP

[8] The credit terms for IDA borrowing has changed to IDA Blend, from IDA Regular in the previous DSA, reflecting the change in Bangladesh's status to Gap.

for domestic debt with maturity of 1–3 year, 10.5 percent for domestic debt with maturity above 3 years and up to 7 years, and 11 percent for domestic debt with maturity above 7 years in the near term. Over the longer term, as monetary policy normalizes, reliance on NSCs declines and domestic bond markets deepen, the cost of domestic debt is expected to marginally decline with 6 percent for T-bills, 7.5 percent for domestic debt with maturity of 1–3 year, 8 percent for domestic debt with maturity above 3 years and up to 7 years, and 8.5 percent for domestic debt with maturity above 7 years in the near term. Average real interest rate on domestic debt is 2.9 percent in FY34. The debt is assumed to be skewed toward T-bonds, with the share of T-bonds issuance with 4-year and above maturity increasing from 50 percent in the medium-term to 75 percent by FY44. As a result, debt service from domestic new debt will significantly increase over the projection period. Lack of household savings and capital market development could pose risk to a shortfall of additional domestic debt absorption, requiring higher for longer domestic interest rates or higher share of external debt.

7. Drivers of debt dynamic is expected to remain broadly unchanged (Figures 3). Historically, PPG external debt has been driven by favorable growth and a positive CA balance. The historical residual is high due to infrastructure-related spending increases in external debt occurring at the same time that growth was strong and the CA was in surplus. Looking forward, the residual declines due to a slowdown in short-term flows. Unexpected changes in debt is close to median of the distribution across low-income countries (LICs).

8. Realism tools suggest that the macroeconomic projections are consistent with the experience of LICs (Figure 4). Bangladesh's 3-year projected adjustment in the primary balance is 1 percent of GDP. This is below the median of the sample of 3-year fiscal adjustments for LICs since 1990 that were under an IMF supported program. Real GDP growth projections under different fiscal multiplier is comparable to the baseline projections. There is high uncertainty of long-term growth projections and, consequently the real GDP growth could be in the range of 6.5 to 7 percent.[9] Public and private investment rates are comparable to the previous DSA, although there was a deviation in FY23 resulting from an investment compression amid elevated costs.

D. Country Classification and Determination of Scenario Stress Tests

9. The debt carrying capacity measured by the Composite Index (CI) for Bangladesh remains at medium, based on IMF's October 2023 World Economic Outlook (WEO) and World Bank's 2022 Country Policy and Institutional Assessment (CPIA). The CI is based on a weighted average of several factors such as the country's real GDP growth, remittances, international reserves, world growth, and the World Bank CPIA score. The CI is calculated for the last two WEO vintages, in this case the October 2022 and April 2023 WEO vintages, and uses 10-year averages of the variables, with 5 years of historical data and 5 years of projections. The threshold for a medium classification is a CI score above 2.69 and below 3.05. Under medium classification, the threshold for the overall

[9] Additional analysis suggests that long-term growth at 6.5 percent does not change external and overall debt risk ratings.

debt carrying is 55 percent of GDP. The threshold for the PV of external debt-to-GDP ratio is 40 percent (Text Table 5).

Text Table 5. Bangladesh: Country Classification

Debt Carrying Capacity	Medium

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintage
Medium	Medium 2.86	Medium 2.87	Medium 2.88

Applicable thresholds

APPLICABLE	
EXTERNAL debt burden thresholds	
PV of debt in % of	
Exports	180
GDP	40
Debt service in % of	
Exports	15
Revenue	18

APPLICABLE	
TOTAL public debt benchmark	
PV of total public debt in percent of GDP	55

10. The calibration of stress tests is similar to the previous DSA. The calibrations of the contingent liability shock is based on the default values for the SOE debt (2 percent of GDP) and financial market component (5 percent of GDP) since they are sufficient to represent potential fiscal risks. A natural disaster stress test was also applied, calibrated at the default setting of a one-time 10 percent of GDP shock to the external debt-to-GDP ratio.[10] The default setting to calibrate for the export shock could exaggerate the adverse impact, as the standard deviation (SD) of export growth over the past 10 years is much higher at 12.7 due to the large fluctuation of exports under a series of external shocks since the pandemic, in contrast to the SD of 4.5 over FY12–FY19.

E. External Debt Sustainability

11. All external debt indicators are on a declining trend and remain below their respective thresholds under the baseline (Figure 1). External PPG debt-to-GDP ratio are expected to settle at around 8 percent by FY44. Under the baseline, all but one (the PV of debt-to-export) sustainability

[10] A World Bank (2016) study, Bangladesh: Building Resilience to Climate Change, estimated that that with a per capita GDP of about US$1,220, the economic losses in Bangladesh over the past 40 years were an about US$12 billion, depressing GDP annually by 0.5 to 1 percentage point.

indicators improve compared to the previous DSA. The PVs of debt-to-exports ratios path is at a higher level compared to the previous DSA. The most extreme shock to the PV of external PPG debt-to-GDP ratio is a one-time depreciation shock. The PV of debt-to-exports breaches the threshold under the most extreme shock, an export shock. However, this temporary two-year breach (starting in FY25) of a small magnitude (lower than previous DSA) is discounted via judgement, as in the previous DSA, in which the breach is driven by large export fluctuations during the pandemic and the post-pandemic recovery periods.[11] Both debt service-to-exports and debt service-to-revenue ratios are on a declining trend and remain under the threshold under the most extreme shock of an export shock and a one-time depreciation shock respectively. Reflecting increases in projected interest payments, debt service-to-revenue ratio for FY24 is higher than the previous DSA, taking it close to the threshold including under baseline. Given the low share of external debt in financing mix, the projected increase in total debt service-to-revenue ratio could raise external debt rollover risks going forward.

F. Public Debt Sustainability

12. Overall public debt indicators suggest a low overall risk of debt distress (Figure 2). The PV of total public debt-to-GDP, while higher compared to the previous DSA, is below its indicative threshold. The largest shock to this indicator is the natural disaster shock. The shock is kept at default calibrations and is equivalent to a one-time 10 percent of GDP shock. All sustainability indicators are on a higher level than the previous DSA, with debt service-to-revenue ratio increasing to 87.7 percent in FY34 and over 100 percent for most years under the most extreme shock, reflecting increase in the cost of domestic debt. This reiterates the importance of raising the revenue-to-GDP ratio in Bangladesh. Increasing the revenue-to-GDP ratio beyond the IMF-supported program period will be critical in providing non-debt financing to growth-enhancing and climate-resilient infrastructure projects. Reforms to improve investment climate remain crucial for attracting FDI.

G. Assessment

13. Bangladesh has a low risk of external and overall debt distress. All but one external debt indicators are below their corresponding thresholds under the most extreme shock, despite an initial increase due to large and ongoing Taka depreciation since FY22. The PV of debt-to-exports breaches the threshold under the most extreme shock to exports. However, a short-lived and small breach (lower than previous DSA), as well as favorable debt dynamics with a declining external debt-to-GDP ratio path, supports the use of judgment and deviation from mechanical rating. The PV of public debt-to-GDP is also below its indicative threshold and is projected to gradually increase to 41.8 percent of GDP in the long term (FY34), reflecting increasing reliance on less concessional domestic financing. The increasing debt service to revenue ratio highlights the urgency of

[11] Specifically, the breach is driven by how a sharp rebound in exports in FY22 affect standard deviation of export growth. Sensitivity analyses, with different standard deviations of export growth (i.e. standard deviation excluding just FY22 or the post-pandemic period or both pandemic and post-pandemic periods), confirm that the PV of debt-to-exports remain below the threshold under an export shock.

mobilizing tax revenue to support much-needed spending to achieve pro-poor, green growth recovery. Development of domestic debt and capital markets is an important component of the IMF-supported program. Reforms to attract FDI, increase domestic revenue mobilization, enhance public investment management, build capacity for public private partnerships, strengthen the financial sector including addressing banking sector vulnerabilities, develop local capital market, and improve debt management remain essential to support much needed capital investments to support long-term growth. Improving investment climate to attract FDIs and promote investment remains high priority for boosting potential growth. Currently, multiple IDA projects are supporting adaptive delta management and climate resilience building.[12] RSF financing under the IMF-supported program would also help finance priorities identified in the Bangladesh Delta Plan (BDP2100) and the National Adaptation Plan (NAP), crowd in other financing, substitute for more expensive domestic financing, improve public debt dynamics, and reduce the balance of payments pressures from import-intensive climate investment.

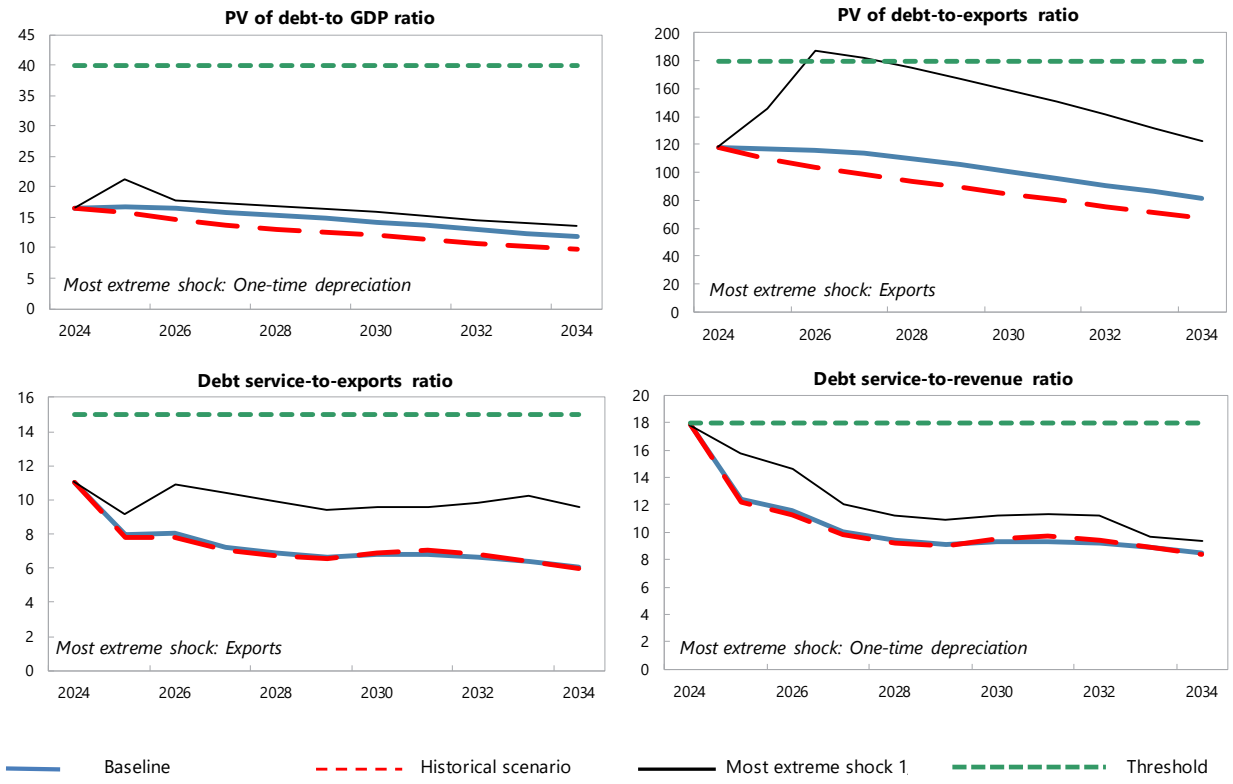
H. Authorities' Views

14. The authorities agreed that the risk of external debt distress and overall risk of debt distress remain low. The authorities also agreed that rising interest rates, both external and domestic, will remain a critical challenge in the coming years, and they have revised up their interest payment projections since the previous DSA. While they restated their commitment to public financial and debt management to improve the debt dynamics and ensure fiscal and debt sustainability, they acknowledged the urgent need to accelerate domestic revenue mobilization to meet financing needs. The authorities reiterated their concern about lowering of the debt carrying capacity, which occurred as part of the 2021 DSA, mostly due to the change in CPIA score. They also reconfirmed that there are no plans to issue Eurobonds.

[12] See, Bangladesh Country Climate and Development Report, for details. World Bank Group. 2022. Bangladesh Country Climate and Development Report. CCDR Series;. World Bank Group, Washington, DC. © World Bank Group. https://openknowledge.worldbank.org/handle/10986/38181 License: CC BY-NC-ND.

Figure 1. Bangladesh: Indicators of Public and Publicly Guaranteed External Debt, FY24–FY34[1/]

(In percent, unless otherwise mentioned)



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	No	
Natural disaster	No	No
Commodity price	n.a.	n.a.
Market financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.0%	2.0%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	21	21
Avg. grace period	6	6

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

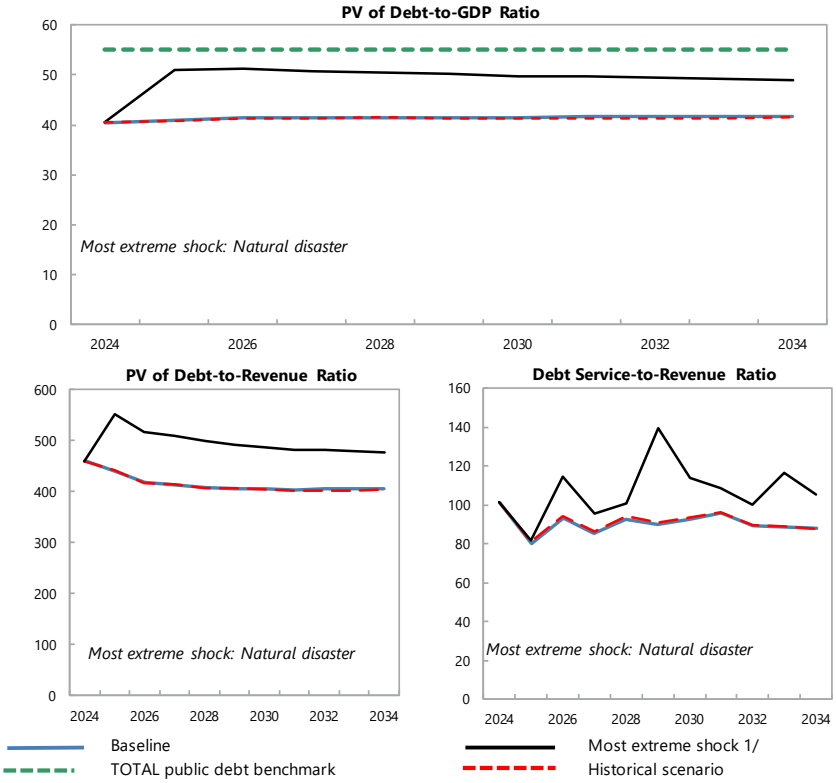
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2034. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

department.

Figure 2. Bangladesh: Indicators of Public Debt, FY24–FY34[1/]

(In percent)



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	18%	18%
Domestic medium and long-term	68%	68%
Domestic short-term	15%	15%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.0%	2.0%
Avg. maturity (incl. grace period)	21	21
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	3.1%	3.1%
Avg. maturity (incl. grace period)	4	4
Avg. grace period	3	3
Domestic short-term debt		
Avg. real interest rate	1.1%	1.1%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

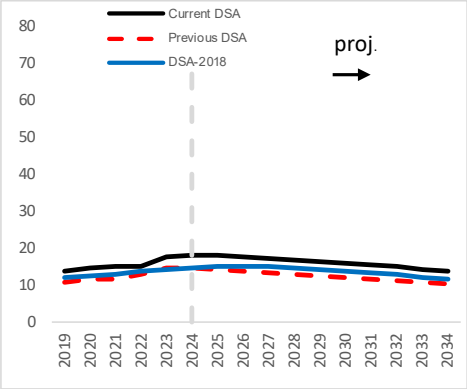
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2034. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

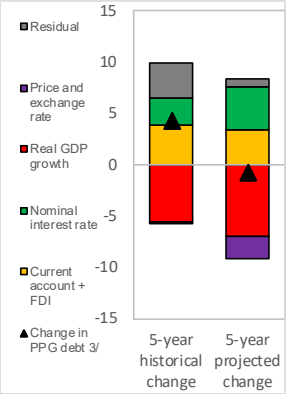
Figure 3. Bangladesh: Drivers of Debt Dynamics – Baseline Scenario

External Debt

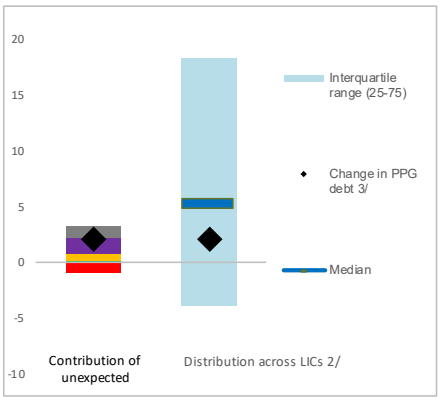
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

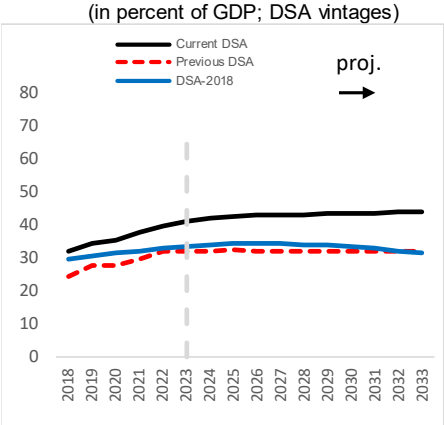


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

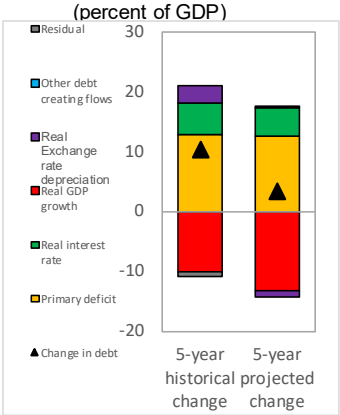


Public Debt

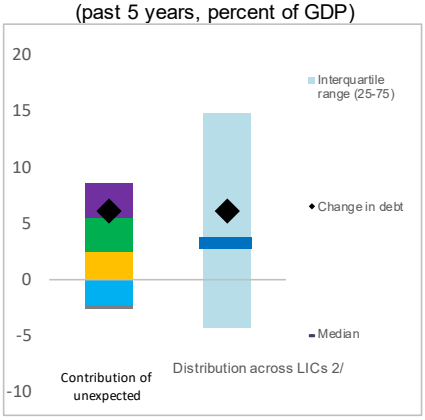
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



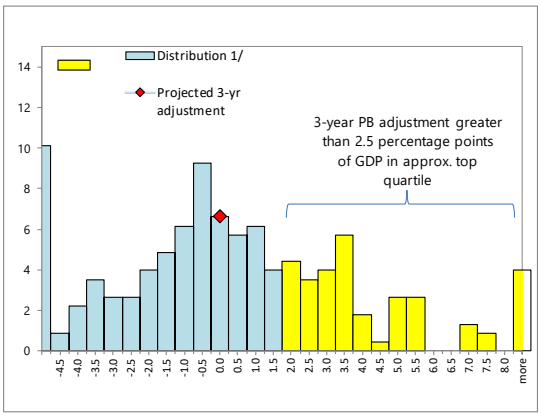
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

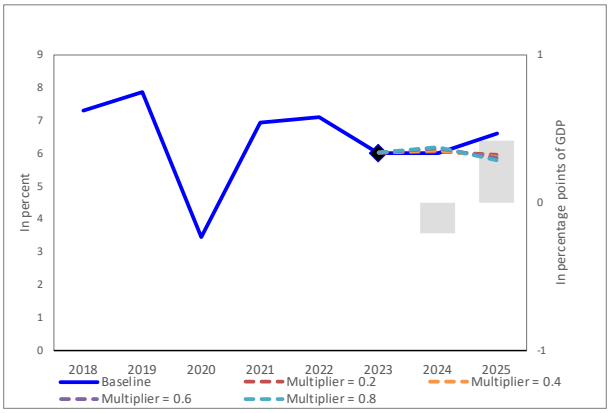
Figure 4. Bangladesh: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



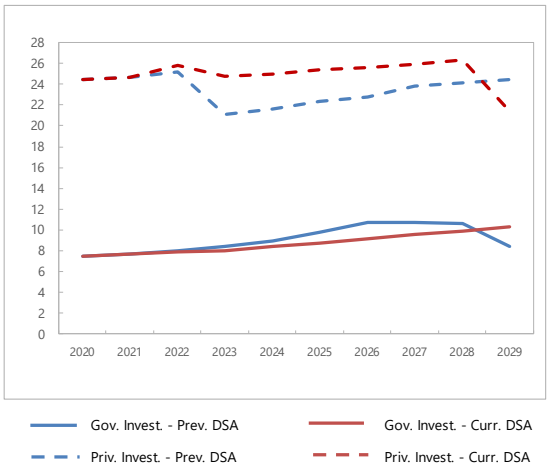
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

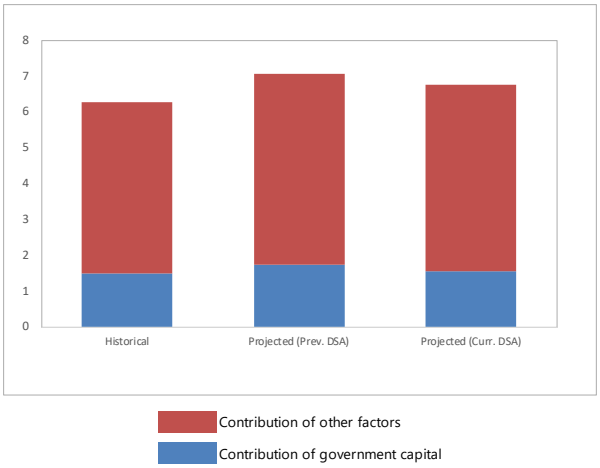


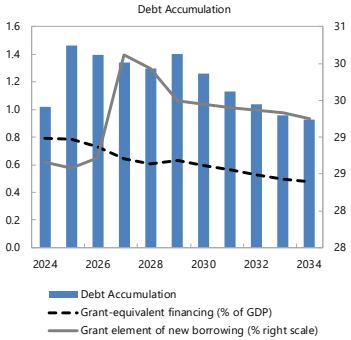
Table 1. Bangladesh: External Debt Sustainability Framework, Baseline Scenario, FY21–FY44 1/

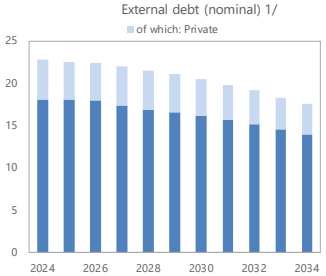
(In percent of GDP, unless otherwise indicated)

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2034	2044	Historical	Projections
External debt (nominal) 1/	**19.6**	**20.8**	**22.7**	**22.8**	**22.5**	**22.4**	**22.0**	**21.5**	**21.1**	**17.5**	**10.5**	**17.9**	**20.7**
of which: public and publicly guaranteed (PPG)	**15.1**	**15.4**	**17.7**	**18.1**	**18.1**	**17.9**	**17.4**	**16.9**	**16.6**	**14.0**	**8.6**	**14.0**	**16.4**
Change in external debt	1.2	1.2	1.8	0.2	-0.3	-0.1	-0.4	-0.6	-0.4	-0.8	-0.6		
Identified net debt-creating flows	**-1.1**	**1.8**	**1.0**	**-0.9**	**0.4**	**0.5**	**0.3**	**0.3**	**0.1**	**0.1**	**1.0**	**-1.1**	**0.1**
Non-interest current account deficit	**0.8**	**3.4**	**-0.1**	**0.0**	**1.9**	**2.2**	**2.2**	**2.2**	**1.8**	**1.7**	**2.6**	**0.4**	**1.7**
Deficit in balance of goods and services	6.4	8.1	4.8	4.9	6.7	6.8	6.6	6.6	6.1	5.5	4.7	**5.5**	**6.0**
Exports	10.7	12.9	13.6	14.0	14.3	14.2	14.0	13.9	14.0	14.5	15.1		
Imports	17.1	20.9	18.4	18.9	21.0	20.9	20.6	20.4	20.1	19.9	19.9		
Net current transfers (negative = inflow)	-6.1	-4.7	-5.0	-4.9	-4.8	-4.6	-4.4	-4.4	-4.2	-3.6	-2.0	**-5.5**	**-4.2**
of which: official	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other current account flows (negative = net inflow)	0.4	0.1	0.1	0.0	0.0	0.1	0.0	0.0	-0.1	-0.2	-0.1	**0.4**	**-0.1**
Net FDI (negative = inflow)	**-0.3**	**-0.4**	**-0.4**	**-0.4**	**-0.9**	**-1.1**	**-1.3**	**-1.3**	**-1.2**	**-1.2**	**-1.2**	**-0.5**	**-1.1**
Endogenous debt dynamics 2/	**-1.5**	**-1.3**	**1.5**	**-0.5**	**-0.5**	**-0.6**	**-0.6**	**-0.5**	**-0.5**	**-0.5**	**-0.3**		
Contribution from nominal interest rate	0.3	0.6	0.9	0.8	0.9	0.9	0.8	0.8	0.8	0.7	0.4		
Contribution from real GDP growth	-1.1	-1.3	-1.3	-1.3	-1.4	-1.4	-1.4	-1.4	-1.3	-1.1	-0.7		
Contribution from price and exchange rate changes	-0.7	-0.6	1.9	...	...	...	...	...	...	...	...		
Residual 3/	**2.3**	**-0.5**	**0.8**	**1.1**	**-0.7**	**-0.6**	**-0.7**	**-0.9**	**-0.5**	**-0.8**	**-1.7**	**1.9**	**-0.6**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**15.9**	**16.6**	**16.8**	**16.4**	**15.8**	**15.2**	**14.7**	**11.8**	**6.8**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**116.9**	**118.2**	**117.2**	**115.7**	**113.4**	**109.7**	**105.3**	**81.4**	**45.1**		
PPG debt service-to-exports ratio	**4.4**	**5.1**	**6.1**	**11.1**	**8.0**	**8.0**	**7.2**	**6.9**	**6.6**	**6.0**	**4.1**		
PPG debt service-to-revenue ratio	**5.0**	**7.5**	**10.0**	**17.8**	**12.4**	**11.6**	**10.1**	**9.4**	**9.1**	**8.5**	**6.0**		
Gross external financing need (Million of U.S. dollars)	16951.5	35283.2	25905.5	26195.2	28011.6	29451.2	30755.4	34235.7	35357.4	51384.2	147472.5		
Key macroeconomic assumptions													
Real GDP growth (in percent)	6.9	7.1	6.0	6.0	6.6	7.1	7.2	7.0	7.0	7.0	7.0	**6.5**	**6.9**
GDP deflator in US dollar terms (change in percent)	4.1	3.2	-8.5	-3.8	0.9	3.4	4.6	5.3	5.3	5.4	5.4	**2.9**	**3.9**
Effective interest rate (percent) 4/	2.1	3.4	4.0	3.8	4.1	4.2	4.1	4.2	4.3	4.2	3.7	**2.8**	**4.2**
Growth of exports of G&S (US dollar terms, in percent)	14.2	33.4	2.5	5.2	9.8	9.7	10.3	11.8	13.8	13.7	13.1	**8.2**	**11.7**
Growth of imports of G&S (US dollar terms, in percent)	18.6	35.5	-14.8	5.0	19.2	10.5	10.0	12.0	11.0	12.7	12.7	**8.5**	**11.9**
Grant element of new public sector borrowing (in percent)	...	...	...	28.7	28.6	28.7	30.1	29.9	29.5	29.3	29.0	**...**	**29.3**
Government revenues (excluding grants, in percent of GDP)	9.3	8.9	8.2	8.7	9.2	9.8	10.0	10.2	10.2	10.3	10.4	**8.5**	**9.9**
Aid flows (in Million of US dollars) 5/	7672	10080	8965	2545	2600	2568	1850	1850	1750	1750	1756		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	0.8	0.8	0.7	0.6	0.6	0.6	0.5	0.3	**...**	**0.6**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	31.4	31.5	31.0	30.4	30.2	29.7	29.4	29.1	**...**	**30.2**
Nominal GDP (Million of US dollars)	416,265	460,201	446,349	455,162	489,538	541,999	607,578	684,881	772,018	1,406,118	4,666,362		
Nominal dollar GDP growth	11.3	10.6	-3.0	2.0	7.6	10.7	12.1	12.7	12.7	12.7	12.7	**9.6**	**11.0**
Memorandum items:													
PV of external debt 7/	...	...	20.9	21.3	21.2	20.9	20.5	19.8	19.2	15.3	8.7		
In percent of exports	...	...	153.6	151.8	148.2	147.2	146.8	142.9	137.4	106.0	57.5		
Total external debt service-to-exports ratio	31.3	32.1	42.7	40.4	33.1	30.6	29.7	29.8	28.3	21.6	11.9		
PV of PPG external debt (in Million of US dollars)			70944	75488	82153	88976	96235	104119	113729	165627	318975		
(PVt-PVt-1)/GDPt-1 (in percent)				1.0	1.5	1.4	1.3	1.3	1.4	0.9	0.4		
Non-interest current account deficit that stabilizes debt ratio	-0.5	2.2	-1.9	-0.2	2.2	2.3	2.6	2.7	2.2	2.5	3.2		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

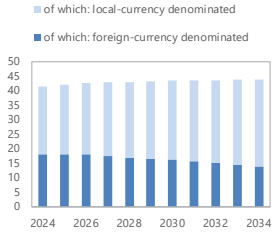
Table 2. Bangladesh: Public Sector Debt Sustainability Framework, Baseline Scenario, FY21–FY44

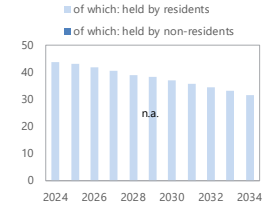
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2034	2044	Historical	Projections
Public sector debt 1/	**35.6**	**37.9**	**39.8**	**41.4**	**42.0**	**42.7**	**42.9**	**43.1**	**43.3**	**44.0**	**44.4**	**32.2**	**43.1**
of which: external debt	**15.1**	**15.4**	**17.7**	**18.1**	**18.1**	**17.9**	**17.4**	**16.9**	**16.6**	**14.0**	**8.6**	**14.0**	**16.4**
Change in public sector debt	1.1	2.2	1.9	1.6	0.6	0.8	0.1	0.2	0.2	0.1	0.0		
Identified debt-creating flows	**0.7**	**1.6**	**2.6**	**0.9**	**0.8**	**1.0**	**0.7**	**0.8**	**0.7**	**0.6**	**0.3**	**1.4**	**0.7**
Primary deficit	**1.6**	**2.1**	**2.6**	**2.8**	**2.3**	**2.5**	**2.5**	**2.5**	**2.6**	**2.5**	**2.2**	**2.2**	**2.6**
Revenue and grants	9.4	8.9	8.3	8.8	9.3	9.9	10.0	10.2	10.2	10.3	10.4	**8.6**	**10.0**
of which: grants	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0		
Primary (noninterest) expenditure	11.0	11.1	10.9	11.6	11.6	12.4	12.5	12.7	12.8	12.8	12.6	**10.8**	**12.5**
Automatic debt dynamics	**-0.9**	**-0.5**	**0.0**	**-1.9**	**-1.5**	**-1.6**	**-1.8**	**-1.8**	**-1.9**	**-2.0**	**-2.0**		
Contribution from interest rate/growth differential	-0.6	-2.1	-1.9	-1.9	-1.5	-1.6	-1.8	-1.8	-1.9	-2.0	-2.0		
of which: contribution from average real interest rate	1.6	0.3	0.3	0.3	1.0	1.2	1.1	1.0	1.0	0.9	0.9		
of which: contribution from real GDP growth	-2.2	-2.4	-2.2	-2.3	-2.6	-2.8	-2.9	-2.8	-2.8	-2.9	-2.9		
Contribution from real exchange rate depreciation	-0.2	1.6	1.9	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**0.4**	**0.6**	**-0.7**	**0.8**	**-0.2**	**-0.2**	**-0.6**	**-0.5**	**-0.5**	**-0.5**	**-0.3**	**-0.2**	**-0.3**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**39.1**	**40.5**	**40.8**	**41.4**	**41.3**	**41.4**	**41.5**	**41.8**	**42.6**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**470.5**	**460.2**	**440.6**	**418.0**	**413.8**	**407.5**	**406.2**	**405.8**	**410.7**		
Debt service-to-revenue and grants ratio 3/	**58.7**	**72.0**	**71.8**	**101.1**	**80.4**	**93.4**	**85.6**	**92.5**	**89.9**	**87.7**	**87.5**		
Gross financing need 4/	7.4	9.0	9.0	11.7	9.8	11.8	11.0	11.9	11.7	11.6	11.3		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	6.9	7.1	6.0	6.0	6.6	7.1	7.2	7.0	7.0	7.0	7.0	**6.5**	**6.9**
Average nominal interest rate on external debt (in percent)	0.0	1.4	2.0	2.2	2.7	2.8	2.7	2.7	2.7	2.7	2.3	**1.0**	**2.7**
Average real interest rate on domestic debt (in percent)	6.1	4.8	3.6	2.0	3.9	4.5	4.2	3.8	3.5	2.9	2.7	**5.3**	**3.4**
Real exchange rate depreciation (in percent, + indicates depreciation)	-1.8	11.6	13.6	...	...	...	...	...	...	...	...	**0.8**	**...**
Inflation rate (GDP deflator, in percent)	4.1	5.0	5.4	6.4	5.5	5.2	5.5	5.3	5.3	5.4	5.4	**5.1**	**5.5**
Growth of real primary spending (deflated by GDP deflator, in percent)	2.3	7.9	4.2	12.8	7.0	14.8	7.6	8.9	7.6	7.1	6.8	**7.0**	**8.6**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	0.5	-0.1	0.6	1.2	1.8	1.8	2.3	2.3	2.4	2.4	2.3	**0.3**	**2.2**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, central bank, government-guaranteed debt . Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Bangladesh: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, FY24–FY34

(In percent)

	Projections 1/										
	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
PV of debt-to GDP ratio											
Baseline	17	17	16	16	15	15	14	14	13	12	12
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2024-2034 2/	17	16	15	14	13	12	12	11	11	10	10
B. Bound Tests											
B1. Real GDP growth	17	17	17	17	16	15	15	14	14	13	12
B2. Primary balance	17	17	17	16	15	15	14	14	13	13	12
B3. Exports	17	18	20	19	19	18	17	16	15	14	14
B4. Other flows 3/	17	18	19	18	17	16	16	15	14	13	13
B5. Depreciation	17	21	18	17	17	16	16	15	15	14	14
B6. Combination of B1-B5	17	19	19	18	18	17	16	16	15	14	13
C. Tailored Tests											
C1. Combined contingent liabilities	17	18	17	17	16	16	15	15	14	14	13
C2. Natural disaster	17	18	18	17	17	17	16	16	15	15	14
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	118	117	116	113	110	105	101	96	91	86	81
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2024-2034 2/	118	110	103	99	94	89	85	80	75	71	67
B. Bound Tests											
B1. Real GDP growth	118	117	116	113	110	105	101	96	91	86	81
B2. Primary balance	118	118	117	115	111	107	103	98	93	88	83
B3. Exports	118	145	**187**	**182**	175	167	159	150	141	132	123
B4. Other flows 3/	118	125	131	128	123	118	112	106	100	94	88
B5. Depreciation	118	117	100	98	96	92	89	85	81	77	74
B6. Combination of B1-B5	118	138	126	143	138	132	126	120	113	106	100
C. Tailored Tests											
C1. Combined contingent liabilities	118	123	122	120	116	114	110	105	100	95	91
C2. Natural disaster	118	130	130	128	124	123	119	113	108	104	99
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	11	8	8	7	7	7	7	7	7	6	6
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2024-2034 2/	11	8	8	7	7	7	7	7	7	6	6
B. Bound Tests											
B1. Real GDP growth	11	8	8	7	7	7	7	7	7	6	6
B2. Primary balance	11	8	8	7	7	7	7	7	7	6	6
B3. Exports	11	9	11	10	10	9	10	10	10	10	10
B4. Other flows 3/	11	8	8	8	7	7	7	7	7	7	7
B5. Depreciation	11	8	8	7	7	6	6	7	6	6	5
B6. Combination of B1-B5	11	9	10	9	8	8	8	8	8	8	8
C. Tailored Tests											
C1. Combined contingent liabilities	11	8	8	7	7	7	7	7	7	7	6
C2. Natural disaster	11	8	9	8	7	7	7	7	7	7	7
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	18	12	12	10	9	9	9	9	9	9	8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2024-2034 2/	18	12	11	10	9	9	9	10	9	9	8
B. Bound Tests											
B1. Real GDP growth	18	13	12	11	10	10	10	10	10	9	9
B2. Primary balance	18	12	12	10	9	9	9	9	9	9	9
B3. Exports	18	12	12	11	10	10	10	10	10	11	10
B4. Other flows 3/	18	12	12	11	10	9	10	10	10	10	10
B5. Depreciation	18	16	15	12	11	11	11	11	11	10	9
B6. Combination of B1-B5	18	13	13	11	10	10	10	10	11	10	10
C. Tailored Tests											
C1. Combined contingent liabilities	18	12	12	10	10	9	10	10	9	9	9
C2. Natural disaster	18	12	12	10	10	9	10	10	10	9	9
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Bangladesh: Sensitivity Analysis for Key Indicators for Public Debt, FY24–FY34

(In percent)

	Projections 1/										
	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
PV of Debt-to-GDP Ratio											
Baseline	40	41	41	41	41	41	41	42	42	42	42
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2024-2034 2/	40	41	41	41	41	41	41	41	41	41	41
B. Bound Tests											
B1. Real GDP growth	40	42	44	44	45	45	46	46	47	47	47
B2. Primary balance	40	42	43	43	43	43	43	43	43	43	43
B3. Exports	40	42	45	45	45	44	44	44	44	44	43
B4. Other flows 3/	40	42	44	43	43	43	43	43	43	43	43
B5. Depreciation	40	43	42	41	40	39	39	38	37	37	37
B6. Combination of B1-B5	40	40	41	40	40	40	40	40	40	40	41
C. Tailored Tests											
C1. Combined contingent liabilities	40	47	48	47	47	47	46	46	46	46	46
C2. Natural disaster	40	51	51	51	51	50	50	50	49	49	49
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
TOTAL public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	460	441	418	414	408	406	405	404	405	405	406
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2024-2034 2/	460	441	418	415	408	406	404	402	403	403	403
B. Bound Tests											
B1. Real GDP growth	460	451	444	444	442	444	446	448	453	456	459
B2. Primary balance	460	449	434	428	421	419	417	415	416	415	415
B3. Exports	460	455	456	448	439	435	432	428	427	424	422
B4. Other flows 3/	460	452	440	434	426	423	421	418	418	416	415
B5. Depreciation	460	465	428	413	397	387	378	370	365	360	355
B6. Combination of B1-B5	460	431	413	405	398	397	395	392	393	393	394
C. Tailored Tests											
C1. Combined contingent liabilities	460	512	481	474	463	458	454	449	448	446	443
C2. Natural disaster	460	551	517	509	498	492	487	482	481	478	476
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	101	80	93	86	92	90	93	96	89	89	88
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2024-2034 2/	101	81	94	86	94	91	93	96	90	89	87
B. Bound Tests											
B1. Real GDP growth	101	82	98	91	98	97	101	105	99	99	99
B2. Primary balance	101	80	95	88	94	94	98	98	91	92	91
B3. Exports	101	80	94	86	93	91	93	96	90	91	89
B4. Other flows 3/	101	80	94	86	93	90	93	96	90	90	89
B5. Depreciation	101	76	90	82	88	86	87	91	85	84	83
B6. Combination of B1-B5	101	78	91	84	90	88	93	93	87	87	86
C. Tailored Tests											
C1. Combined contingent liabilities	101	80	107	92	97	123	106	103	95	106	98
C2. Natural disaster	101	82	114	96	101	139	114	108	100	116	105
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.